Circular Wireless RF Systems with Recyclable and Degradable Components Based on Additively-Manufactured Liquid Metal Interconnects

Xiaochuan Fang, and Mahmoud Wagih

Abstract— **Fabricating sustainable circuits with low-waste end-of-life is crucial for enabling an electronics circular economy. Here, we present the first circularity-driven fabrication flow for sustainable wireless circuits based on recyclable liquid metals and degradable dielectrics. The proposed recyclable radio frequency (RF) circuits consist of 3D-printed substrates with fluidic channels, e.g. using polylactic acid (PLA), filled with Galinstan interconnects. At their end-of-life, the ICs and liquid metal interconnects are recovered and re-used in functional circuits, over multiple iterations. The printed dielectrics are degraded, with under 2% by weight waste produced. Various RF circuits are demonstrated, up to 4 GHz, including transmission lines based on Coplanar Waveguides (CPWs), with high-performance DC/RF control and filtering, and broadband high-efficiency radiation in a CPW antenna. Active microwave systems are then demonstrated including a voltage-controlled oscillator (VCO), and a multi-layered design for a broadband power amplifier (PA). The circuits maintain a stable performance across over 9 metal and 3 IC re-use cycles. The reliability of the circuits is verified by continuously operating months-old circuits for over 30 hours as well as under-water storage, demonstrating their reliability. Our results pave the way for multi-function, green wireless circuits, reducing waste electrical and electronic equipment (WEEE) and allowing component circularity.**

Index Terms—**3D printing, biodegradable electronics, electronic waste, green electronics, liquid metal, radio frequency, recyclable electronics, sustainable electronics.**

I. INTRODUCTION

WITH the rapid development of pervasive wireless applications, namely for internet of things (IoT) networks, an enormous number of electronic devices are manufactured. Consequently, a significant amount of waste electrical and electronic equipment (WEEE) is generated at the end of these devices' life cycles [1]-[2]. Currently, the recycling rate of WEEE is very low. Instead, much of this WEEE is typically discarded, for grinding and incineration, leading to substantial metal pollution in soil and water [3].

Various precious and scarce metals are used in electronics manufacturing, which are rarely recycled at end-of-life [4]-[9]. The highest carbon footprint and materials consumption is attributed to Integrated circuits (ICs), essential in any radio frequency (RF) circuit. In order to minimize the emissions and critical rare metals (CRMs) consumption of these ICs, simple, low-temperature, and hazardous chemicals-free recycling is fundamental to enabling a circular economy in electronics [10]-[11].

Sustainable, recyclable or biodegradable attracted significant interest [12]-[22]. Although some recyclable circuits have been reported recently, they face several limitations. The practicality of fabricating these devices remains a critical challenge. Many recyclable circuits require complex fabrication processes that consume significant amounts of energy or water. Additionally, these processes often rely on highly specialized instruments or specifically prepared materials [12]-[18]. Furthermore, approaches involving chemical molding or thermal decomposition can damage materials or devices, presenting substantial challenges. As a result, the reliability and practicality of RF circuits produced using these methods are often questioned. Currently, most biodegradable circuits are based on biodegradable conductors such as zinc (Zn) or magnesium (Mg). However, integrating these metals onto a printed circuit board (PCB) for circuit fabrication can be difficult [19]-[22]. Additionally, biodegradable circuits have yet to be reported for RF applications.

Galinstan-based liquid metal has been widely adopted in the construction of electronic circuits and devices due to its outstanding characteristics, including non-toxicity, flexibility, stretchability, and deformability [23]-[29]. Its flexibility allows it to be shaped into specific circuits or components as needed and fully removed when no longer required, making liquid metal recyclable in room temperature. Additionally, the conductivity of Galinstan (EGaIN) is 3.4×10^{6} S/m, which is close to the conventional raw metals, ensuring the performance of the circuits [30]. Nevertheless, liquid metal has yet to be applied in recyclable RF electronics or devices. To explain, existing RF circuits and devices based on liquid metal are composed of conventional metals and non-recyclable dielectrics, making them difficult to recycle [31]-[33]. On the other hand, while some recyclable devices have been reported, they are designed for digital current (DC) or terminal applications, where only simple line resistances were considered, implying that further design and characterization

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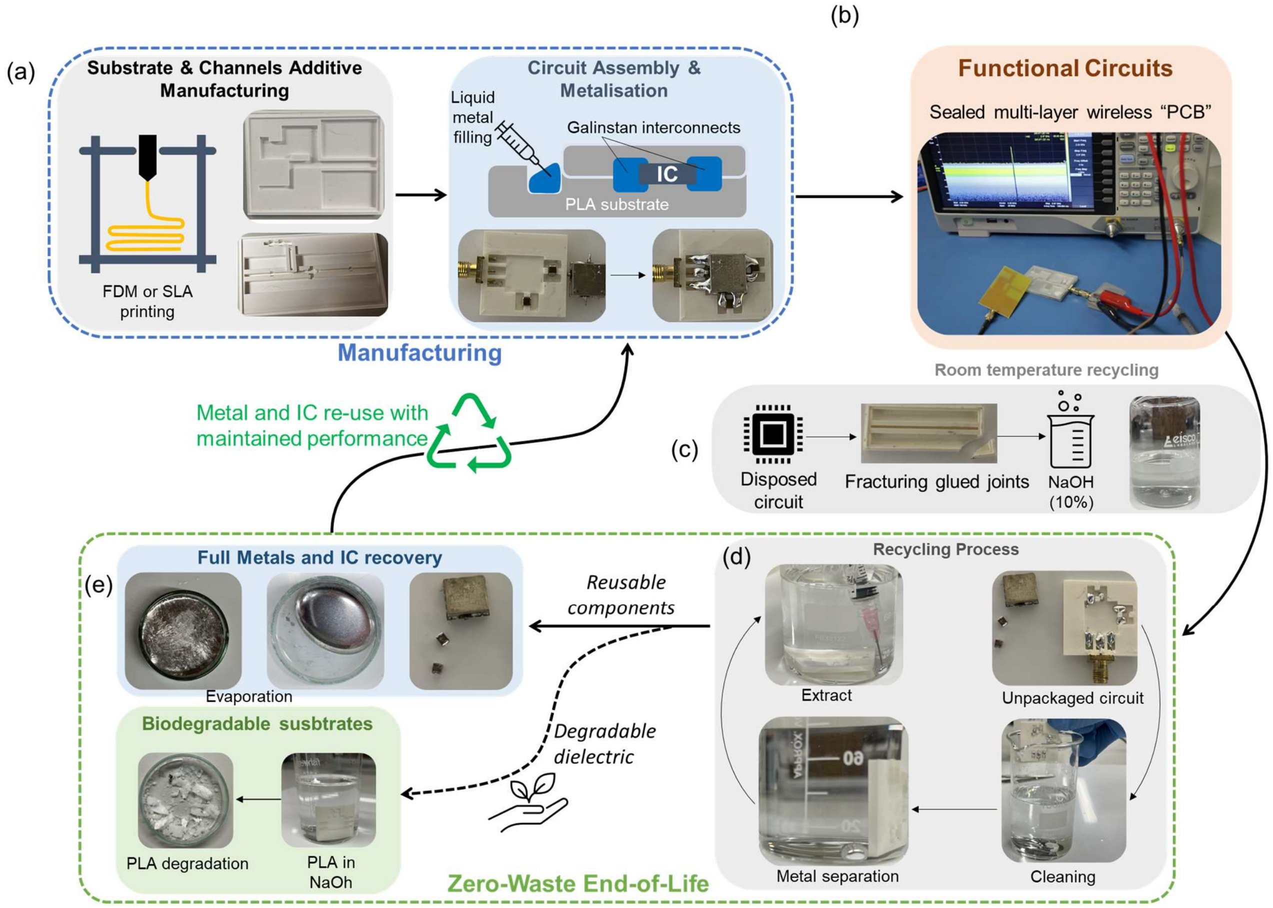

Fig. 1. The proposed circular fabrication flow for 3D wireless circuits: (a) Fabrication stage, combining additive manufacturing and liquid metal filling; (b) utilization stage, achieving similar performance to FR4 PCBs; (c) end-of-life and circuit disposal; (d) room temperature IC and liquid metal retrieval process; (e) zero-waste end-of-life, showing the degradation of the substrate and full components and metal recovery for re-use in new circuits.

is needed prior to supporting RF and wireless applications [34]-[36].

In this paper, we propose the first set of low-waste, recyclable and biodegradable, RF circuits based on recyclable Galinstan-based liquid metal and degradable 3D-printed dielectrics. The design we proposed is the only liquid metal circuit system used for RF applications that consists entirely of recyclable and biodegradable components. The fabrication and recycling process was verified across up to 9 cycles, for passive RF interconnects. Four circuit demonstrators are then shown over 3 recycling runs, including, RF/DC-filtering lines, an active voltage-controlled oscillator (VCO), a broadband CPW antenna, and a broadband multi-layered power amplifier. All circuits achieving stable and consistent performance.

II. Circuit Fabrication and Recycling Process

Fig. 1 shows the proposed full life cycle-driven low-waste fabrication flow, and how it enables an electronics circular economy. It should be noted that the liquid metal and the ICs are recycled, while the substrate and fluidic channels are biodegraded. The first step in the fabrication process is the additive manufacturing of the substrates and fluidic channels, i.e. analogous to the PCB's substrate. This can be achieved using either fused deposition modeling (FDM) or stereolithography (SLA) printing. The material used for 3D-printing the substrates or channels is polylactic acid (PLA), which is biodegradable. The circuits are then assembled by placing the ICs in their designated positions. Subsequently, liquid metal is injected using a syringe to interconnect the ICs, as seen in Fig. 1(a). For the circuit requires large-in-fill liquid metal, i.e. monopole antenna, we adopt an approach based on an overfill-and-draw-out method. This involves injecting excess liquid metal into the channel until it is completely filled, then draw out the liquid metal inside the channel and make it flat. This prevents the liquid metal from leakage after the circuit is sealed.

Once the circuits are fabricated, they are sealed, making them operational. As seen in our 3D-printed prototypes, gaps have been intentionally left around the edges of the substrates and fluidic channels. Simultaneously, ridges have been incorporated around the lid of the prototypes in corresponding positions. This design allows the lid to initially fit securely onto the prototype. Additionally, we applied super glue around the corners of the prototypes, specifically in areas relatively far from the channels containing the liquid metal, ensuring a proper seal. The sealed and functional circuit is illustrated in Fig.1(b), and throughout the paper for the different

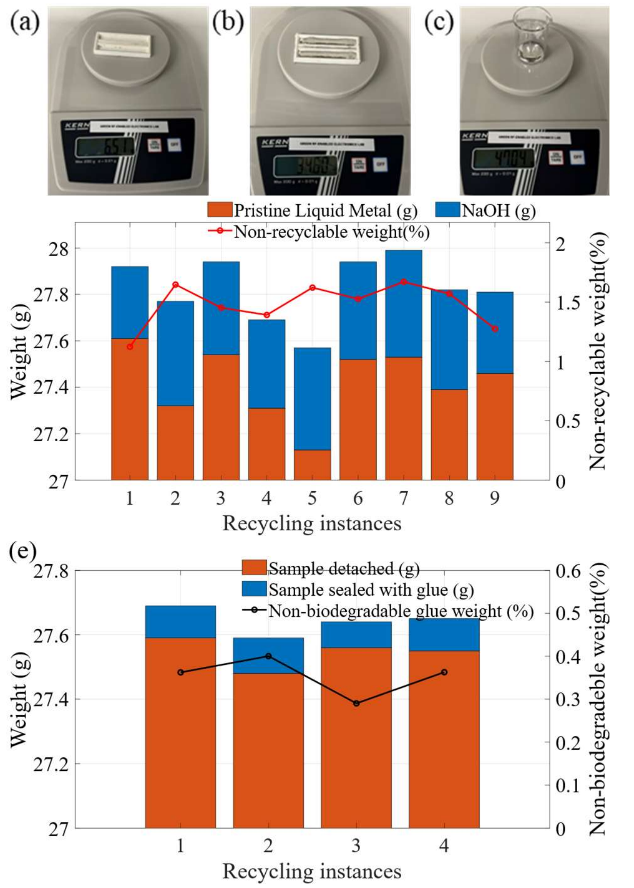

Fig. 2. Circuit waste rate weighting: (a)weighting empty channel, (b) weighting pristine liquid metal; (c) weighting recycled liquid metal (d) waste weight produced by NaOH; (e) waste weight produced by glue.

components. This process could be applied to more complex multi-layered designs, where overlapping traces are needed.

At the end of their first life, the disposed circuits undergo recycling and degradation. For this process, a 10% sodium hydroxide (NaOH) solution is prepared, as in Fig. 1(c). The first step is de-attaching the circuits. The disposal circuits will be manually fractured at the edges where the super glue was applied. We use this approach particularly when the circuits contain ICs that are vulnerable and could be damaged or contaminated by NaOH. This method allows us to recover the ICs or other components before immersing the circuits in the NaOH solution.

Once the model is de-attached, they could be recycled and degraded by putting into the NaOH solution. The NaOH breaks down the oxidized surface of the liquid metal, causing it to detach from the circuits and shrink into small droplets that are easily separated from the substrate and fluidic channels [37]. To accelerate the speed of the recycling, a small amount of solution that contains both liquid metal and NaOH is then extracted from the beaker. While the extracted liquid metal is mixed with the NaOH solution initially, the significant difference in density between the two allows the NaOH to float on top of the liquid metal. Once the NaOH evaporates, low-contamination liquid metal is left behind, which can then

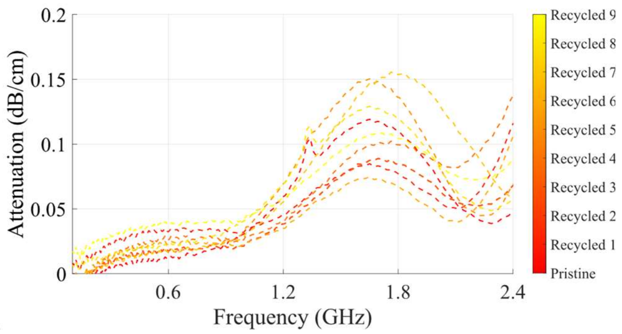

Fig. 3. Repeatability (n=9) of the measured liquid metal based lossy transmission line representing by attenuation.

TABLE I
SPECIFICATIONS OF THE PROPOSED CIRCUIT PROCESS

Specifications	Performance
Maximum Dimensions	330×240x330 mm
Minimum Dimensions (tolerance)	1×1 mm (±10%)
Height	1.5 mm
Hole diameter	0.6 mm
Minimum trace/spacing	1 mm
Trace thickness	1 mm
Waste weight	<2%

be used to fabricate new circuits. As for the degradable polymers used in the 3D-printed substrates, those can be left to degrade. In our process, the PLA is directly degraded in the NaOH solution. The recovered ICs and metals can then be re-introduced into new 3D-printed molds, allowing the precious and high-value metals to be re-used over multiple iterations.

The waste from the sample primarily comes from the non-recyclable NaOH. To calculate the waste weight of the proposed sample, we measured the weight of the sample before and after the recycling process. Fig. 2 illustrates the weighing process used to assess waste. First, the emptied 3D-printed channel was weighed. Then, pristine liquid metal was added to the channel, allowing us to measure and calculate its weight. After recycling, the liquid metal was transferred into a pre-weighed beaker, and the weight of the recycled liquid metal was recorded. The difference in weight between the pristine and recycled liquid metal represents the waste produced by NaOH. Fig. 2(d) presents the weight of the pristine and recycled liquid metal, as well as the waste weight after nine recycling cycles. The waste weight was calculated to be less than 1.6%. Additionally, the sealing and detachment processes may contribute to waste due to the use of super glue. The waste weight from these processes was also measured and is shown in Fig. 2 (e), with a calculated value of less than 0.4%. Therefore, the overall waste rate is less than 2%.

The overall cost of the proposed circuit is primary arise from contamination and unrecyclable consumables, which is under £1.1 per each circuit.

To verify the recyclability of the proposed process, a liquid metal transmission line was fabricated and recycled nine times.

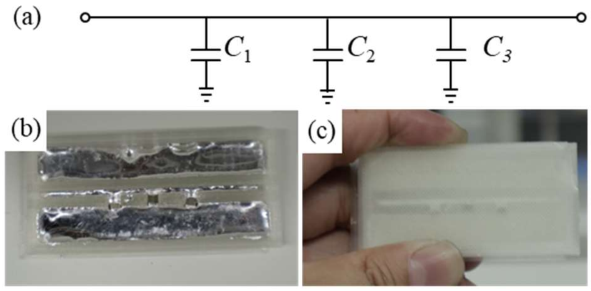

Fig. 4. RF-coupled DC line: (a) schematic of passive DC line; (b) assembled passive DC-line with liquid metal interconnects; (c) sealed passive DC-line.

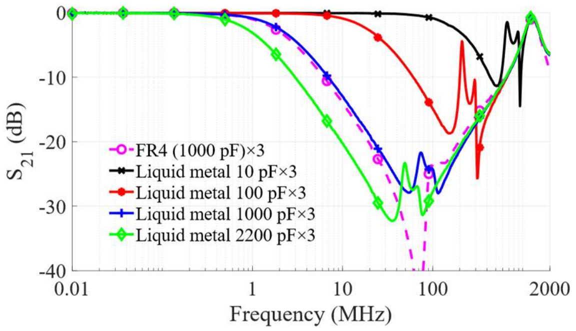

Fig. 5. Transmission Coefficient of RF-decoupled DC feed line with different capacitor.

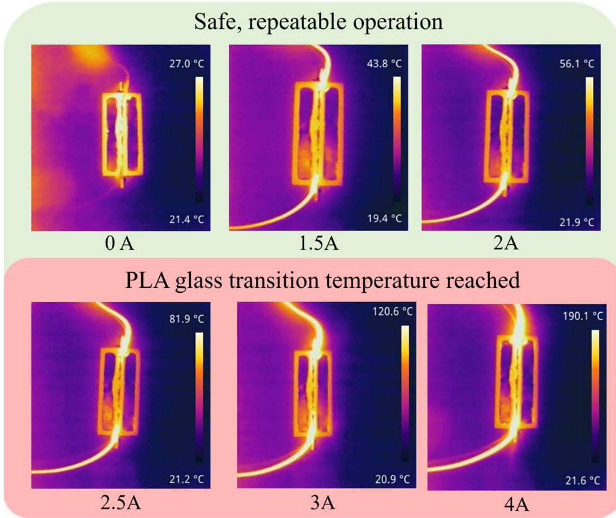

Fig. 6. IR pictures showing the current handling of a power line fabricated using the proposed process.

The attenuation of the transmission line is shown in Fig. 3, over all 9 cycles. The frequency range in Fig. 3 up to 2.4 GHz, as the SMA connector introduces an inductive mismatch above 2.5 GHz. The differences in attenuation among the recycled transmission lines are less than 0.1 dB/cm. Fig. 3 thus demonstrates the effectiveness of the proposed circuit fabrication and recycling process.

Table I provides the key specifications of the proposed fabrication process. The proposed fabrication process secures the circuit with low-profile and large area to be fabricated. Nevertheless, the main limitation of the proposed process arises from the resolution of the 3D printer. This means that a design with a complex structure can still be realized if it meets the current resolution limitations or if the 3D layout is strategically arranged, as demonstrated in our power amplifier circuit. On the other hand, numerous studies have demonstrated that liquid metal can be injected into microfluidic channels with extremely small dimensions [38], i.e. only 5 µm, which is smaller than the resolution achievable using conventional PCB manufacturing methods. This means that the injection of liquid metal should not influence the capability of the proposed circuit.

III. Circular and Low-Waste Circuit Demonstrators

In this section, we demonstrate four circular RF circuits, showcasing the scalability and performance of the proposed circuit fabrication flow. The performance of all four circuits has been verified to remain almost intact after undergoing one or more recycling passes, validating the circularity promise of the proposed flow.

A. Passive RF and DC Lines

The first structures considered are RF-decoupled DC interconnects. Fig. 4 (a) shows the schematic of a line with 3 decoupling capacitors. In this circuit, the ground planes and the transmission line are fully connected between the two ports. Three gaps are introduced between the transmission line and the ground planes, where three surface-mounted 1206 capacitors can be placed. The use of three different capacitors allows for better differentiation in the transmission coefficient among various capacitance values. Additionally, using three shunt capacitors improves the quality factor, enabling a wider stopband to be achieved. The impedance of a capacitor is given by $Z_c = 1/j\omega C$, where the ω is the frequency of the signal; he cut-off frequency of the decoupled supply line is therefore dependent on the capacitance value of the surface-mounted capacitor [39].

Fig. 4 (b) and (c) presents the assembled and sealed prototype of the proposed DC feed line. This design features two liquid metal ground planes and a middle-gapped transmission line. The gap size between the transmission line and the ground, as well as the width of the transmission line, has been calculated to achieve a characteristic impedance of approximately 50 Ω. The calculated line width is 1.5 mm, and the gap width is 2.7m. The key feature of a DC supply line is rejecting high frequency signals, acting as a low-pass filter (LPF), especially for kHz to MHz noise from nearby electromagnetic interference (EMI) sources.

The circularity of the proposed RF coupled DC feed line is achieved by recycling the liquid metal interconnects and refilling them into newly printed PLA fluidic channels. It is worth noting that the surface-mounted capacitor has high resilience and is minimally affected even after resoldering. As a result, the recycling of the capacitor is not considered in the proposed DC feed line. However, one of the capacitors used in this line will also serve as the blocking capacitor in the DC-blocking transmission line, which will be discussed later.

The transmission coefficient of the DC feed line is shown in Fig. 5, for varying decoupling capacitor values. The

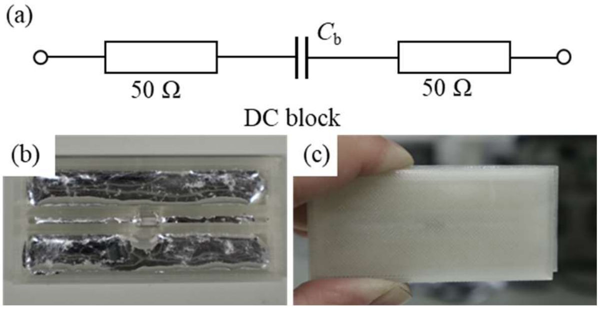

Fig. 7. DC-blocking RF transmission line implementation with liquid metal interconnect: (a)schematic; (b) assembled DC-blocking line; (c)Sealed DC-blocking line.

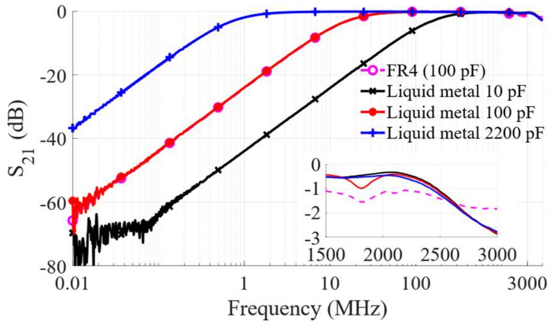

Fig.8. DC-blocking transmission line with different capacitors: broadband S_{21} measured from DC to 3 GHz.

performance closely matches the rigid FR4 reference, showing the suitability of the proposed fabrication for filtered DC lines.

The power handling of the proposed circuits was then investigating for high currents. A DC power supply was used to drive the liquid metal/PLA line with input current ranging from 0 A to 4 A. The temperature was measured at different current levels using an infrared camera. Fig. 6 shows the measured temperature profile at different current levels. For inputs under 2 A, the temperature increase is within the operation range of PLA. At an input current of 2.5 A, the circuit's temperature rises to 82 °C, exceeding the glass transition temperature of PLA. Beyond 3 A, the temperature increase causes the solvents in the PLA filament to evaporate.

Fig. 7 (a) presents the schematic of the DC-blocking transmission line. In this line the blocking capacitor is placed at the middle gap to connect the transmission lines on both sides. The capacitor used in the DC-blocking transmission line is recycled from aforementioned DC feed line. Fig. 7 (b) and (c) presents the assembled and sealed DC-blocking RF line with liquid metal interconnects. The circular fabrication process for the DC-blocking RF transmission line is similar to that of the DC feed line.

Fig. 8 compares the transmission coefficient of the DC-blocking line using surface-mounted capacitors with different capacitances. It is evident that larger capacitance values result in lower cutoff frequencies. The results also demonstrate that the circular process introduced in the previous section ensures the proposed DC-blocking transmission line maintains its desired performance.

The performance of the liquid metal-based DC-blocking transmission line is also compared with that of a hardwired transmission line made from FR4. Notably, the gap and line

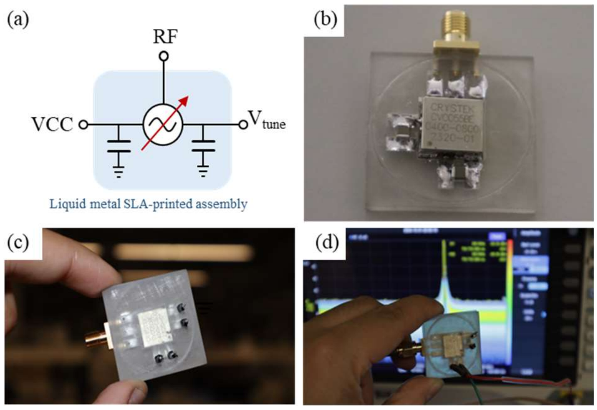

Fig. 9. VCO circuit with liquid metal interconnects: (a) schematic; (b) assembled unpackaged circuit; (c) sealed VCO circuit;(d) measurement setup.

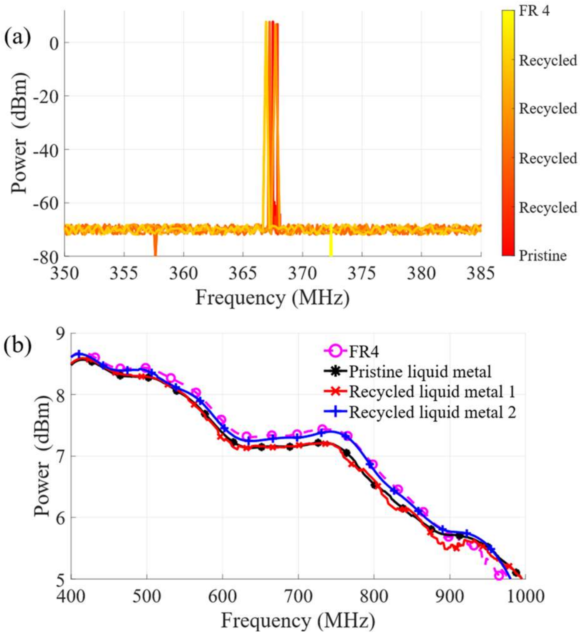

Fig. 10. Measured VCO results for pristine and recycled liquid metal and ICs, closely matching the performance of a pristine FR4: (a) measured power output of the VCO in the liquid metal process; (b) measured RF power output of the VCO across and its tuning range.

width of the FR4 transmission line are adjusted to achieve a 50 Ω characteristic impedance. The comparison shows that when the same capacitance is used, the DC-blocking curves of the liquid metal and FR4 transmission lines perfectly overlap. This indicates that the conductivity of the liquid metal supports acceptable performance for the proposed DC-blocking circuit. The inset figure in Fig. 8 shows the S_{21} of the proposed circuit across the RF frequency range. This figure indicates that the liquid metal-based DC-blocking line is more lossy compared to the FR4-based DC-blocking line. This is because the thickness of the liquid metal line in the proposed circuit makes it more difficult to achieve impedance matching compared to the FR4-based transmission line.

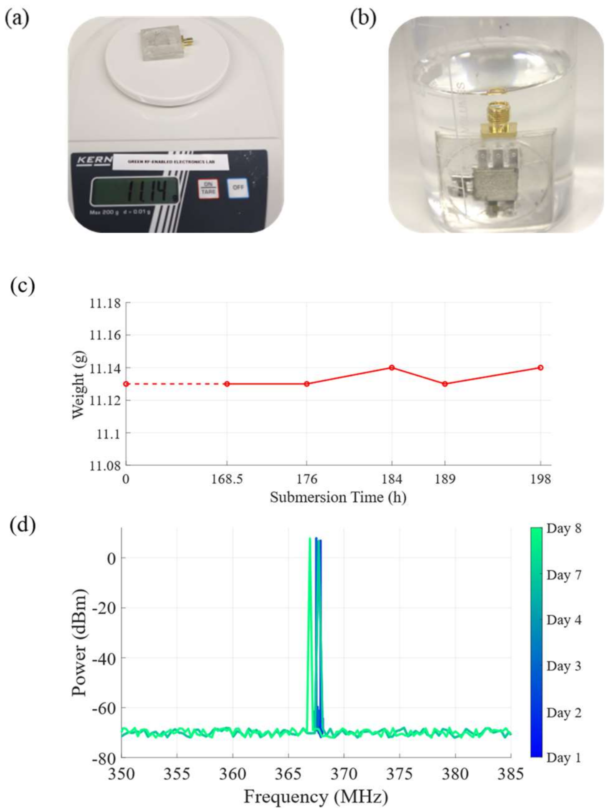

Fig. 11. VCO reliability measurement: (a) weighting the sealed VCO circuit; (b) submerge the VCO circuit in the water; (c)weight of the VCO circuit after it was submerged in the water and dried five times in 30 hours after sealed for 7 days; (d) measured power output of VCO over a week of immersion in water (dried before measurement).

B. RF voltage-controlled oscillator (VCO)

To demonstrate an active RF system, a VCO circuit is implemented based on an off-the-shelf component. The VCO used in this circuit is the CVCO55BE-0400-0800, chosen for being available for different frequencies in the same package, simplifying testing across frequencies, and for its wide tuning range.

The RF output of the VCO was measured using a spectrum analyzer, before the voltage is varied, across its tuning range. A reference circuit was also fabricated on FR4 and measured under the same conditions; the datasheet indicates an output power of 5 dBm. Fig. 9 (a) shows the schematic of the VCO circuit. It includes two DC ports: the DC supply voltage (VCC) and the DC tuning voltage (V_{tune}). The RF port of the VCO circuit is connected to an SMA connector using liquid metal interconnects, enabling RF power output. Fig. 9 (b) shows the assembled unpacked VCO circuit with liquid metal interconnects.

Following recycling, the stability and reliability of the VCO is evaluated based on the power output over the VCO's tuning range, measured using a spectrum analyzer. After measuring the pristine sample, the VCO circuit was disposed of through

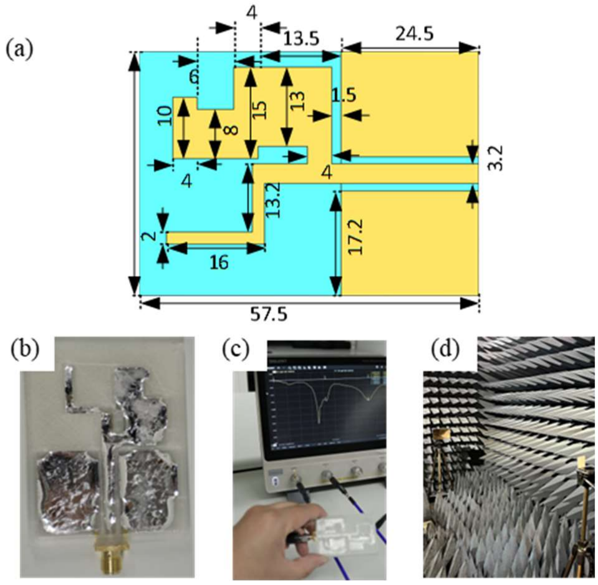

Fig. 12. Broadband CPW liquid metal antenna implementation: (a) layout and dimensions of the monopole (unit: mm); (b) assembled monopole antenna; the sealed antenna during measurements on (c) VNA and in (d) anechoic chamber.

the propose recycling process, shown in Fig. 1(d). The recovered IC, SMD passives, and liquid metal were used to fabricate a new sample, in a freshly 3D-printed package.

As shown in Fig. 10 (a), the recycling process was repeated four times demonstrating the repeatability of the fabrication process, and feasibility of recycling RF components using the proposed flow. The output power differences across these configurations are less than 0.4 dB. The variations observed in the frequency are due to the device's free-running nature, where the oscillation is sensitive to e.g. temperature changes. However, the power stability indicates that neither the metal nor ICs have degraded through the recycling process.

The repeatability of the VCO circuit's performance after recycling is further demonstrated by comparing the power levels of the VCO across its full operating band, as shown in Fig. 10 (b). As seen in Fig. 10 (b), the operating band of the proposed VCO circuit extends from 400 MHz to 1000 MHz, the specified tuning range. Fig. 10 (b) also shows that the broadband power output generated by a fresh VCO with pristine liquid metal interconnects closely matches that of recycled liquid metal interconnects and a VCO with copper interconnects on an FR4 PCB, even after 2 recycling procedures. Thus, indicating that the brief NaOH exposure does not degrade the IC's performance. The maximum variation in output power is 0.2 dB. The results in Figs. indicate that the proposed VCO circuit achieves circularity with performance comparable to that of a conventional PCB-mounted VCO circuit.

The stability of the sealed VCO circuit was investigated, demonstrating that the liquid metal cannot leak from the sealed interconnects, and that high-moisture environments will not change the electrical performance. To this end, the VCO

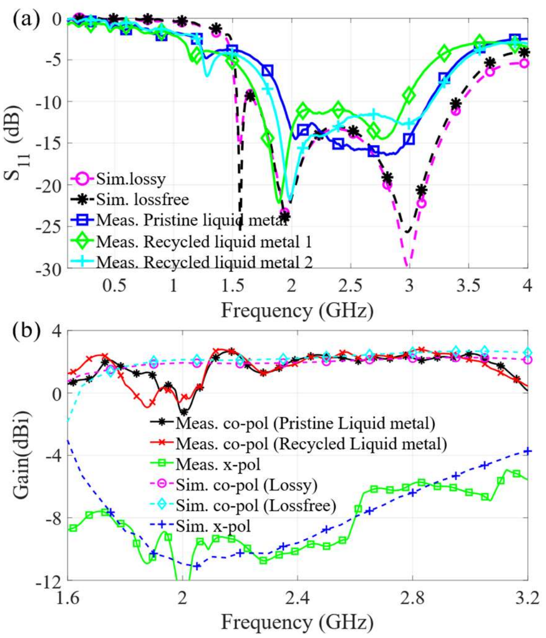

Fig. 13. Antenna performance: (a) Simulated/Measured reflection coefficient for the proposed antenna; (b) boresight peak gain and polarization purity of the proposed antenna.

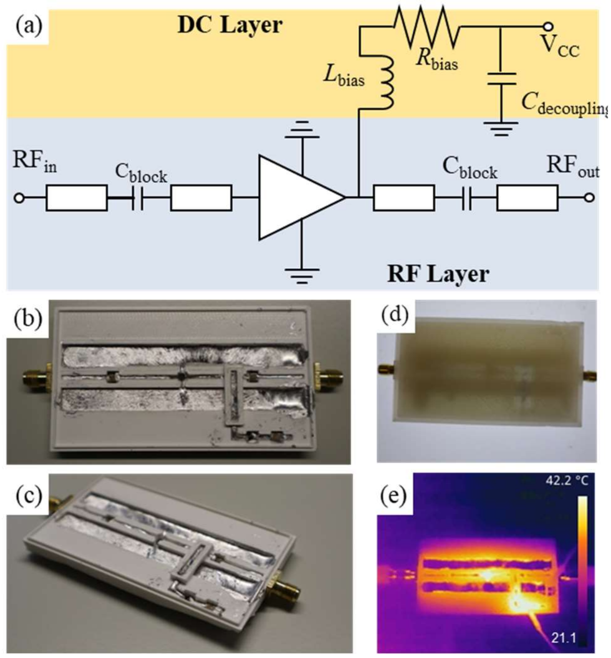

Fig. 14. Recyclable RF power amplifier design: (a) schematic of PA circuit; (b) assembled PA from front view; (c) assembled PA from side view; (d) sealed PA; (e) IR picture showing the hotspots in the circuit.

circuit was submerged in a beaker of tap water, emulating extreme-humidity environments. Fig. 11 (a) and (b) show the measurement procedure, and the circuit as it is immersed and stored under water. After weighing, the VCO circuit was submerged in water, as shown in Fig. 11(b). Prior to the RF measurements, the circuit was dried and weighed, observing under 1% change in its weight. This process was repeated after 0.5 hour, 8 hours, 16 hours, 21 hours, and 30 hours, after 7 days of sealing. Fig. 11 (c) shows the weight of the VCO circuit has a minor variation for less than 0.01g after repeating submerging and drying for five time, indicating no liquid metal leakage during this process.

The output power of the VCO circuits was measured over a week of being submerged in water, demonstrating no leakage in or out of the circuit, and no change in performance. Fig. 11 (b) shows output power level of the VCO circuit after 7 days it was submerged in the water and dried. The output power level of the VCO remained consistent throughout.

From a safety perspective, Galinstan liquid metal used in the proposed circuits is biosafe [40], meaning that any potential contamination caused by the liquid metal is harmless to biological systems. Galinstan has been widely adopted in various biomedical applications, such as wearable antennas [41] and body sensors [42]–[43]. Moreover, several effective methods exist for cleaning up liquid metal contamination. Among them, alcoholic solutions offer a safe and efficient option, as they are non-corrosive and self-evaporating [44].

C. Broadband CPW Antenna; A Large-In-Fill Design

To demonstrate the feasibility of fabricated components with large metal areas with a non-uniform geometry, a broadband monopole antenna is investigated. Unlike the previously introduced circular circuits, this antenna comprises 2D shapes rather than 1D wire-like interconnects. Filling a 2D shape with liquid metal poses challenges because Galinstan-based liquid metal is a non-Newtonian fluid [45], meaning its surface tension is nonlinear and decreases significantly during surface oxidation. As a result, liquid metal cannot easily fill a bulky 2D-shaped fluidic channel. Various methods exist to address this issue, though many require expensive instruments or stringent procedures [46]. On the other hand, the overfill-and-draw-out process we invented in this paper allows liquid metal to be formed in 2D shape within 10 minutes. The large-in-fill monopole antenna we proposed also capable of measured in an anechoic chamber where it is vertically positioned, and liquid metal inside is not leaked or cross to adjacent channels.

Fig. 12 (a) shows the layout and dimensions of the proposed monopole antenna. The radiator is an asymmetric monopole, which excites metallic slits in different locations, thereby extending the antenna's bandwidth. The asymmetric monopole consists of five rectangular strips of varying sizes. Given the nonlinear surface tension of liquid metal, rectangular fluidic channels are easier to fill than channels with other shapes (e.g., Y-shaped, meander), which are typical for ultra-wideband (UWB)-inspired antennas [47]-[48]. To further reduce the reflection coefficient within the antenna's operating bandwidth, an arm radiator is placed opposite the monopole.

Fig. 12 (b) shows the assembled antenna with liquid metal interconnects. The antenna is simulated using CST Microwave Studio. After sealing, the reflection coefficient is measured using a vector network analyzer (VNA), and the boresight gain is measured in an anechoic chamber, as shown in Fig. 12

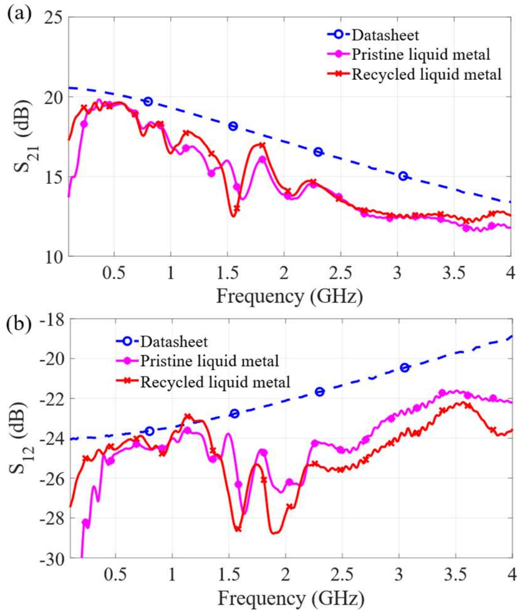

Fig. 15. Measured (a) gain and (b) isolation of the proposed PA circuit compared to the datasheet.

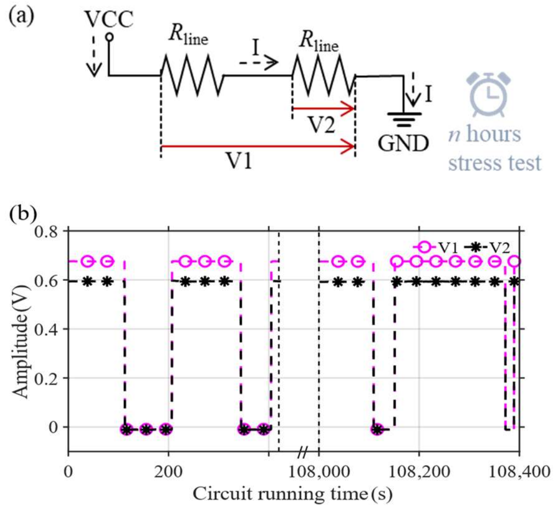

Fig. 16. DC feed line reliability verification: (a) schematic of the verification circuit; (b) DC feed line potential differences with the DC power source that randomly switching on/off, data collecting in 0 mins and 30 hours.

(c) and (d). Since the antenna does not include any ICs or surface-mounted components, the recycling process involves only the recycling of the liquid metal and the degradation of the PLA substrate.

Fig. 13 (a) presents the simulated and measured reflection coefficients of the antenna. The simulated reflection coefficient suggests an impedance bandwidth of 1.7 GHz to 3.43 GHz (67%). The first measurement, taken after filling the fluidic channel with pristine liquid metal, shows an impedance bandwidth of 1.9 GHz to 3.2 GHz (54%). After the circular process, the reused metal antenna's reflection coefficient measurement shows an impedance bandwidth of 1.7 GHz to 2.9 GHz (52%), while a second circular antenna demonstrates an impedance bandwidth of 1.8 GHz to 3.2 GHz (54%). The differences between these results are less than 3%, indicating good repeatability.

Fig. 13 (b) shows the boresight gain of the proposed antenna. The gain is calculated using the free-space path loss method, with a TBMA4 dual-ridge horn as the reference antenna. The difference in boresight gain between pristine and recycled liquid metal antennas is less than 0.5 dB, primarily attributed to fabrication and measurement tolerances. The simulated boresight gain also suggests a reduction of 0.3 dB due to the combined loss of liquid metal and substrate. The cross-polarization of the proposed antenna is more than -8 dB across the 1.7 GHz to 3 GHz range. The measured cross-polarization closely matches the simulated values and remains within the expected range. The differences between the pristine and recycled antennas fall within the ±1 dB uncertainty of such path loss-based two-antenna measurements and demonstrate that the recycled liquid metal mostly retains its RF conductivity.

D. Active Broadband RF System: Power Amplifier

The final circuit considered is a broadband RF power amplifier (PA), which combines both DC and RF design elements, mandating the use of a complex double-layer structure. The PA used in the proposed circuit is the Mini-Circuits ERA-5+, usable from low frequencies up 4 GHz.

Fig. 14 (a) shows the schematic of the proposed PA circuit. The RF input port is connected to the RF power source through a DC-blocking surface-mounted 1206 capacitor (100 μF), as with the RF output. A bias tee is connected to the RF output branch of the PA circuit to supply DC power. The bias tee comprises a 0603-bias inductor (16 nH), a 1206 bias resistor (33.2 Ω), and a 1206 decoupling capacitor (0.1 μF). Per the datasheet, a 7 V DC power supply is used for the PA.

Fig. 14 (b) and (c) shows the assembled model of the proposed PA circuit from front view and side view, respectively. Fig. 14 (d) shows the sealed PA circuit. The PA is interconnected via a CPW with a characteristic impedance of 50 Ω. The circuit includes two grounded pins, and RF input and output ports. It is worth noting that the PA is connected to a single-layer CPW-fed transmission line. This design creates a challenge: if the bias tee is placed on the same layer as the RF branch, the ground plane will be interrupted, leading to discontinuity. The stability of the PA relies heavily on the stability of the RF input and output, meaning the reflection coefficients at both must remain low. Discontinuities in the ground plane can cause abrupt changes in the transmission line's impedance. To address this issue, a portion of the biased tee fluidic channel was printed on a different layer and bridged over the ground plane, enabling the separation of the DC and RF branches. Two liquid metal pins are placed at both ends of the bias tee bridge, allowing the DC supply to connect to the RF output. The IR distribution of the PA is shown in Fig. 16 (e), the heat distribution showing the DC current flows from the PA to the bias tee.

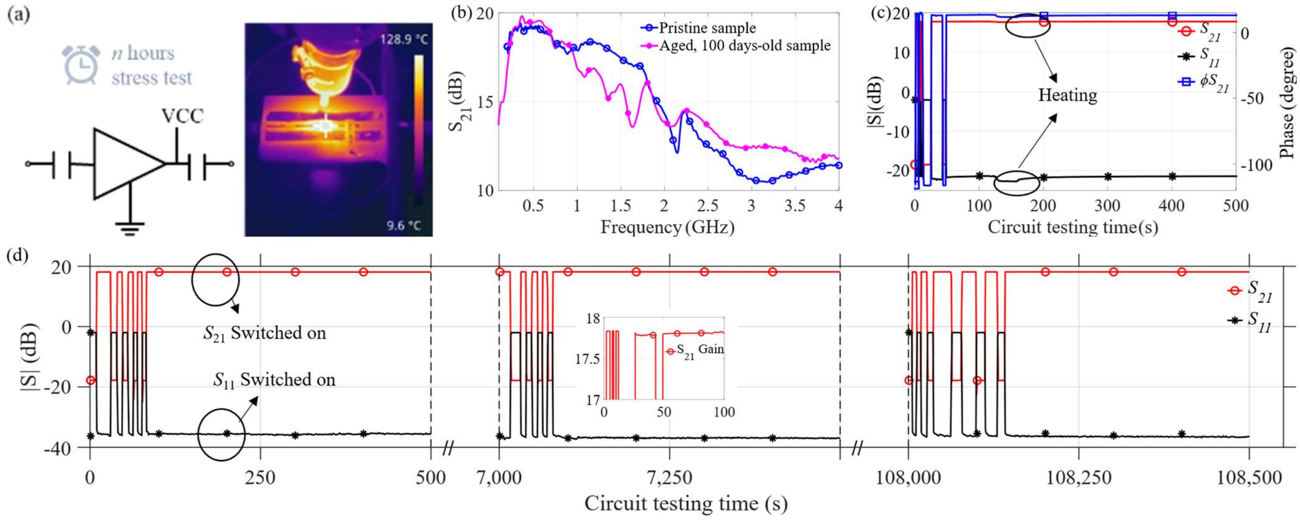

Fig. 17. PA circuit reliability verification: (a) Schematic of verification circuit and IR picture showing external thermal stressing ; (b) S_{21} comparison of a pristine sample, and a sample after three months of storage and movement; (c) S_{21}, S_{11} and S_{21} phase of the PA circuit over time with the external heating, at the start of a 32-hour period; (d) S_{21} and S_{11} of the PA circuit over the time with the DC power source that randomly switching on/off, data collecting in 0 mins, 2 hours and 30 hours with inset figures of maximum gain S_{21} of the PA circuit.

Fig. 15 shows the measured gain and isolation of the liquid metal-assembled PA circuit, with an input RF power of 0 dBm, before and after recycled. The gain measured from the PA circuit with pristine liquid metal interconnects closely overlaps with that measured from the circuit using recycled liquid metal interconnects, confirming the circularity of the proposed PA circuit. The measured gain reaches a maximum of 18 dB at 0.5 GHz and decreases to 13 dB around 3 GHz. The gain drop below 0.5 GHz is due to the used inductance, where a higher-inductance RF block could be used to extend the operation range. When compared with the PA datasheet, the measured gain of the circuit exhibits under 1.5 dB of additional insertion losses. This discrepancy is primarily attributed to impedance mismatches from uneven heights of the ground planes, transmission lines, and the connections between the SMA connectors and the liquid metal interconnects. This explanation is supported by the significantly increased gain loss observed beyond 2 GHz, which aligns with the attenuation trend shown in Fig. 5, for a passive transmission line. The isolation of the PA, in Fig. 15 (b) closely matches the manufacturer-measured performance, indicating that the liquid metal assembly does not cause additional undesired coupling.

IV. CIRCUITS RELIABILITY VERIFICATION AND COMPARISON

A: Circuit Reliability Investigation

To explore the reliability of the proposed circuits, we conducted long-term measurements on the DC feed line and PA circuit. Fig. 16 (a) shows the schematic of the measurement setup for the DC feed line. The DC feed line is considered low-loss and has inherent resistance. The VCC of the DC feed line is supplied by a DC power source. Potential differences were measured at the half-length and full length of the DC feed line using two probes connected to an oscilloscope. These potential differences are denoted as V1 and V2, respectively.

Fig. 16 (b) shows the measured potential differences of V1 and V2 when the DC power supply provides a 2.5 A current. The overall resistance of the DC feed line circuit was calculated to be 0.28 Ω. During this measurement, the DC feed line operated for two consecutive 500 s periods. The potential differences recorded after the circuit has operated for 0 min and 30 hours. Theoretically, V2 should be half of V1 since it measures the half-length of the transmission line. However, as seen in Figure 7, in the proposed circuit, the jumper exhibits a more significant IR drop compared to the DC feed line. Since the proposed DC feed line is highly low loss, when the jumper connected to the negative end of the DC feed line is more lossy than the one connected to the positive end, V2 will be slightly greater than half of V1. During the measurement, the power supply was randomly switched on and off. The amplitude variations indicate that the proposed DC feed circuit is reliable for long-term operation and multiple test cycles.

The reliability of the PA circuit has been verified by operating it for over 30 hours. Fig. 17 (a) shows amplifier circuit, with the IR image showing the external heating applied to demonstrate its stability. Prior to conducting this verification, the liquid metal, ICs, and surface-mounted devices (SMDs) were recycled from a previous PA circuit that last for more than 100 days and used to fabricate a new PA circuit on a newly printed substrate. Fig. 17 (b) shows that the performance of the newly fabricated circuit closely matches that of the previous circuit.

Firstly, we examined the influence of temperature changes on the proposed PA circuit at a fixed frequency (700 MHz) where the PA circuit exhibits a stable S_{21} of 18 dB during

TABLE II
COMPARISON OF RECYCLABLE/BIODEGRADABLE CIRCUITS AND SYSTEMS

Ref.	Recyclable and/or degradable conductor used for circuit	Conductivity (S/m)	Dielectric (Degradable)	IC and metal reuse	Fabrication process	Maximum Frequency tested	RF systems (>1 GHz)
[15]	EG/ChCl, EG/NH4 Ac/C3H8O/ChCl	Less than 120	Woven silicone (No)	No	Filling	10 KHz	No
[16]	HEC/PEDOT:PSS	1×10^4	N/A	No	Evaporation	DC	No
[17]	PEDOT:PSS	1×10^4	PVA (Yes)	No	Heating and condensate	DC	No
[18]	PEDOT:PSS	6.2×10^4	P3HT (No)	No	Patterning and electrolytic	10 KHz	No
[19]	Zn/Mg	$1.69\times10^7/6.3\times10^5$	Na-CMC/PEO (Yes, conditional)	No	Printing	150 MHz	No
[20]	Ti/Cr/Au	167	POMaC (Yes)	No	Laser cutting	10MHz	No
[21]	Mo	1.69×10^7	PHBv (Yes)	No	Sputter coated	N/A, DC	No
[34]	Galinstan	3.4×10^6	Paraffin Wax (Yes)	IC (No) Metal (Yes)	Heating, Melting and molding	350 MHz	No
[36]	Printed Silver (non-recyclable)	6.3×10^7	Glass (no)	IC (Yes), Metal (No)	Dispenser printing	N/A, DC	No
This work	**Galinstan**	**3.4×10^6**	**PLA (Yes)**	**Yes, 3+ (ICs), 9+ (metals)**	**3D printing and filling**	**3 GHz**	**Yes**

operation. To this end, a hot air gun was used to heat the PA IC and its surrounding interconnects and package, as shown in Fig. 17 (a). While temperature variations do affect the S_{21}, S_{11}, as shown in Fig. 17 (c), this is attributed to the variations expected based on the PA's datasheet, and do not accelerate any degradation in the circuit's performance.

The stability of the circuit was demonstrated over a stress test of over 30 hours, shown in in Fig. 17 (d). First, the PA circuit was operated with the s-parameters recorded continuously. At the start of each recording interval, the DC power supply was cycled between the on and off states, showing a repeatable response and peak gain, as seen in the inset of the graph. The data collecting in o min, 2 hours and 30 hours showing the maximum gain of the PA remains. The inset figure showing the PA has a consistent maximum gain of 18 dB. These results indicate that the proposed PA circuit demonstrates strong reliability. The testing and reliability methodology exceeds that reported in the previous works compared in Table II, in duration and depth (e.g. under-water testing, and >30-hour operation tests). Beyond the presented tests, the reliability of the proposed circuit could also be evaluated using established reliability testing methods, such as thermal or power cycling [49]–[51].

B: Comparison with Prior Recyclable/Biodegradable Circuits

Table II compares the proposed recyclable circuits with some typical recyclable and biodegradable circuits based on PCBs. As shown in Table 1, many recyclable circuits utilize polymer conductors, which have relatively low conductivities. Additionally, some of these circuits contain dielectrics that are neither recyclable nor biodegradable or require strict conditions for degradation (which demands significant amounts of energy and water). Moreover, most of these circuits need to be fabricated using approaches that involving chemical molding or thermal decomposition, e.g. evaporation, heating and condensate, electrolytic, etc.. These approaches could potentially damage the circuit, SMDs, or ICs. This may explain why these circuits do not demonstrate the capability for IC recycling. On the other hand, some biodegradable circuits are based on high-conductivity conductors. However, their operating frequencies are limited, and they are not suitable for RF circuits. Works such as [34] and [36] have investigated recycling circuits implemented using Galinstan liquid metal, or printed silver traces. However, these works either do not report IC/SMD recycling or use non-biodegradable substrates. They also require a complicated fabrication process. Most importantly, none of them have been applied to circuits or systems for RF applications (i.e., >1 GHz).

The circular circuit manufacturing approach proposed in this paper is the first to explore IC reuse with a focus on high-performance sub-3 GHz RF applications. The process does not involve chemical molding or thermal decomposition. thus, the liquid metal and ICs used in the proposed circuit can be almost fully recycled while maintaining their performance, at room temperature.

V. Conclusion

We presented the first end-to-end flow for fabricating fully recyclable and low-waste RF circuits based on liquid metal interconnects and 3D-printed dielectrics, enabling a circular economy for wireless circuits and systems. The ICs/surface mount components and the liquid metal interconnects can be fully recycled, whilst the resin/PLA substrate can be degraded, in a low-temperature and low-waste process.

The capabilities and limitations of the process have been defined across medium and high-power RF and DC applications, with stress tests for over 30 hours. Four different circuits, including DC-blocking RF transmission lines, AC-decoupled DC supply lines, an RF VCO circuit, a monopole antenna and a PA based on SMD ICs have been explored. The potential for a circular economy of electronic ICs and interconnects was demonstrated across a number of recycling runs, showing a mostly unchanged electrical response for more than 9 re-use cycles. The proposed flow will enable further work in recyclable, additive, and low-waste electronics manufacturing across wireless and IoT applications and serves as a direct route to reducing emissions and materials consumption in the electronics industry.

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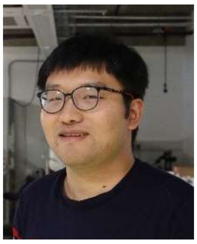

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